

QUIZ 2

5 questions and 1 min for each one

At which level of the Scientific Question Formulation Funnel does a researcher match a specific biological phenomenon to a concrete dataset, data type, and analytical method?

A. Biological phenomenon

B. Data-driven question

C. Operationalized question

D. Testable hypothesis

In observational bioinformatics, the PECO framework is often used instead of PICO. What does the 'E' in PECO specifically represent?

A. Evaluation

B. Evidence

C. Environment

D. Exposure

When constructing a PubMed search strategy, what is the primary function of using the Boolean operator 'OR'?

- A. To exclude specific publication types, such as reviews or clinical trials.
- B. To broaden the search by including synonyms or related terms within a single concept.
- C. To narrow results by ensuring both terms appear in every paper.
- D. To limit the search to papers published within a specific date range.

A student proposes the question: 'Are the microbiomes of lean and obese individuals different?' According to the course material, why is this question considered 'confounded'?

- A. It is a trivial question that has already been answered by thousands of studies.
- B. There is no measurable variable or comparison defined in the question.
- C. It is too broad for a single dataset to answer.
- D. The observed differences might actually be caused by age, diet, or medication rather than obesity itself.

In the context of evaluating source quality, why might a study be considered 'weaker evidence' if it reports only unadjusted p -values?

- A. Unadjusted p -values indicate that the sample size was too small to be significant.
- B. It means the authors used outdated software versions for their RNA-seq pipeline.
- C. Unadjusted p -values mean the study did not use any public data repositories.
- D. Failing to adjust for multiple testing or covariates increases the risk of reporting false-positive associations.